EE1007
Introduction to Computer Science Laboratory

Lab 3
Control Statement I (flow control)

Andy, Yu-Guang Chen
Assistant Professor, Department of EE
National Central University
andyygchen@ee.ncu.edu.tw

Lab 3.1
Sorting (30%)

- In this problem, you need to write a program to sort three numbers
- The program will obtain 3 different integers from keyboard as inputs
- Then the program then will sort the numbers in increasing order
- Finally, the program will output sorting result on your screen
- Following are some examples:
  - Input: 3, -1, 5 ; Output: -1 -> 3 -> 5
  - Input: 7, 8, 4 ; Output: 4 -> 7 -> 8
  - Input: 6, 5, 9 ; Output: 5 -> 6 -> 9
- The range of each integer is $[-2^{31}\sim2^{31}-1]$

Lab 3.1
Sorting (30%)

- The screen dialog should appear as follows:

```
Please enter 3 different integers:
12
56
-4
The sorting result from small to large is -4->12->56
Process returned 0 (0x0)   execution time : 2.909 s
Press any key to continue.
```


Lab 3.2 Factorial Calculation (30%)

- In this program, you need to find factorial for an arbitrary given nonnegative integer
- The program will take a nonnegative integer from keyboard, and the range of the nonnegative integer is less than or equal to 20
- Then the program will output the corresponding factorial on the screen

Lab 3.2 Factorial Calculation (30%)

- The program must "calculate" the factorial instead of just output the results
  - Writing a code in the following way is strictly prohibited

```
    cout << "Please enter a nonnegative interger: ";
    cin >> a;

    if(a==3)
        answer = 6;
    else if(a==5)
        answer = 120;
    else
        answer = 720;

    cout << "The corresponding factorial is " << answer;
}
```

- Following are some examples:
  - Input: 3 ; Output: 6
  - Input: 5 ; Output: 120
  - Input: 6 ; Output: 720

Lab 3.2 Factorial Calculation (30%)

- The screen dialog should appear as follows:

```
Please enter a nonnegative integer: 7
The corresponding factorial is 5040
Process returned 0 (0x0)   execution time : 6.370 s
Press any key to continue.
```

Lab 3.3
Binary to Decimal (40%)

- In this program, you are asked to convert a given binary number to decimal number
- The user will input an unsigned binary number with less than or equal 19 bits from your keyboard
- Then the program needs to convert the unsigned binary number into it's corresponding decimal number and output on the screen
- Following are some examples:
  - Input: 101 ; Output: 5
  - Input: 111 ; Output: 7
  - Input: 1011 ; Output: 11
  - Input: 1111 ; Output: 15
  - Input: 101011 ; Output: 43
  - Input: 111111 ; Output: 63

Lab 3.3
Binary to Decimal (40%)

- The screen dialog should appear as follows:

```
Please input an unsigned binary number: 1101
The corresponding decimal number is 13
Process returned 0 (0x0)   execution time : 13.062 s
Press any key to continue.
```

Evaluation

- Lab3.1(30%)
  - 5 testcases, 6% each
- Lab3.2(30%)
  - 5 testcases, 6% each
- Lab3.3(40%)
  - 5 testcases, 8% each





Andy, Yu-Guang Chen

Assistant Professor, Department of EE, NCU

Email: andyygchen@ee.ncu.edu.tw